Convex Powered–Descent Guidance: Problem 4 (Acikmese & Ploen) in MATLAB/CVX

Companion note for `p4.m`

Abstract

This note documents the optimization problem solved by the script `p4.m` and explains how the MATLAB/CVX code realizes it. The script is a 2D implementation of *Problem 4* from the enhanced convex formulation of the Mars powered–descent guidance problem by Acikmese and co–authors [1, 2]. Given fixed initial and terminal states and a fixed total flight time t_f, the program returns a fuel–optimal thrust profile, formulated as a second–order cone program (SOCP) through a lossless change of variables and a Taylor linearization of the mass–flow bounds.

1 Problem setting

A lander with dry mass m_{dry} and initial wet mass m_{wet} must be steered from an initial state $(\boldsymbol{r}_0, \boldsymbol{v}_0)$ to a desired terminal state $(\boldsymbol{r}_f, \boldsymbol{v}_f)$ in a fixed flight time t_f, while minimizing fuel. The planet is Mars, modeled with a constant gravity vector $\boldsymbol{g}$. The vehicle has n_{eng} identical engines canted by an angle ϕ from the body axis; throttle is bounded between $\eta_{\min}$ and $\eta_{\max}$, and specific impulse I_{sp} is constant.

The numeric values used in `p4.m` are listed in Table 1.

Symbol	Meaning	Value
m_{dry}	dry mass	$1505\,\text{kg}$
m_{wet}	initial wet mass	$1905\,\text{kg}$
I_{sp}	specific impulse	$225\,\text{s}$
g_0	Earth gravity constant	$9.80665\,\text{m/s}^2$
$\boldsymbol{g}$	Mars gravity vector	$(0,\ -3.7114)\,\text{m/s}^2$
$T_{\max}$	max total thrust at full throttle	$6 \times 3100\,\text{N}$
$\eta_{\min}, \eta_{\max}$	throttle limits	$0.3,\ 0.8$
ϕ	thruster cant angle	27°
$\boldsymbol{r}_0$	initial position	$(2000,\ 1500)\,\text{m}$
$\boldsymbol{v}_0$	initial velocity	$(100,\ -75)\,\text{m/s}$
$\boldsymbol{r}_f, \boldsymbol{v}_f$	terminal state	$\mathbf{0},\ \mathbf{0}$
t_f	total flight time	$75\,\text{s}$
Δt	discretization step	$1\,\text{s}$
N	number of knot points	$t_f/\Delta t + 1 = 76$

Table 1: Parameters used in `p4.m`.

2 Continuous–time dynamics and lossless convexification

2.1 Original (non–convex) problem

Let $\boldsymbol{r}(t) \in \mathbb{R}^2$ be position, $\boldsymbol{v}(t) = \dot{\boldsymbol{r}}(t)$ the velocity, $m(t)$ the mass, and $\boldsymbol{T}(t) \in \mathbb{R}^2$ the thrust vector. The translational dynamics are

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \tag{1}$$

$$m\,\dot{\boldsymbol{v}} = \boldsymbol{T} + m\,\boldsymbol{g}, \tag{2}$$

$$\dot{m} = -\alpha\,\|\boldsymbol{T}\|\,, \qquad \alpha := \frac{1}{I_{sp}\,g_0\,\cos\phi}. \tag{3}$$

The thrust magnitude is bounded by effective min/max values obtained by projecting the canted engines onto the body axis,

$$\rho_1 := \eta_{\min}\,T_{\max}\cos\phi, \qquad \rho_2 := \eta_{\max}\,T_{\max}\cos\phi, \qquad \rho_1 \le \|\boldsymbol{T}(t)\| \le \rho_2. \tag{4}$$

Fuel–minimization is equivalent to minimizing $\int_0^{t_f} \|\boldsymbol{T}\|\,\mathrm{d}t$, or equivalently maximizing $m(t_f)$.

This problem is non–convex because of the *lower* bound $\|\boldsymbol{T}\| \ge \rho_1$ (a non–convex annulus) and the division by m in the velocity dynamics.

2.2 Change of variables

Following Acikmese & Ploen, define

$$z(t) := \ln m(t), \qquad \boldsymbol{u}(t) := \frac{\boldsymbol{T}(t)}{m(t)}, \qquad s(t) := \frac{\Gamma(t)}{m(t)}, \qquad \|\boldsymbol{T}(t)\| \le \Gamma(t), \tag{5}$$

where Γ is a scalar slack on thrust magnitude. In the new variables the dynamics become

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \tag{6}$$

$$\dot{\boldsymbol{v}} = \boldsymbol{u} + \boldsymbol{g}, \tag{7}$$

$$\dot{z} = -\alpha\,s, \tag{8}$$

$$\|\boldsymbol{u}\| \le s. \tag{9}$$

Relation (9) is a *second–order cone* constraint and is convex. Acikmese & Ploen show that the relaxation $\|\boldsymbol{T}\| \le \Gamma$ is lossless: at the optimum it holds with equality, so no fuel is wasted by working with Γ instead of $\|\boldsymbol{T}\|$. The lower thrust bound is mapped to a bound on s that depends on mass:

$$\frac{\rho_1}{m(t)} \le s(t) \le \frac{\rho_2}{m(t)}. \tag{10}$$

Equation (10) is still non–convex in (z, s) because $1/m = e^{-z}$ is convex in z. A conservative Taylor expansion around a reference z_0 convexifies it, as described in §3. The objective reduces to

$$\max\ z(t_f) \quad\Longleftrightarrow\quad \max\ m(t_f) \quad\Longleftrightarrow\quad \min\ \text{fuel used}. \tag{11}$$

3 Discretized convex program

The trajectory is discretized on N equally spaced knot points $t_i = (i-1)\Delta t$, $i = 1, \ldots, N$. Decision variables are

$$\boldsymbol{r}_i, \boldsymbol{v}_i, \boldsymbol{u}_i \in \mathbb{R}^2, \qquad z_i, s_i \in \mathbb{R}, \qquad i = 1, \ldots, N.$$

Objective.

$$\max \; z_N. \tag{12}$$

Boundary conditions.

$$\boldsymbol{r}_1 = \boldsymbol{r}_0, \quad \boldsymbol{v}_1 = \boldsymbol{v}_0, \quad z_1 = \ln m_{\text{wet}}, \qquad \boldsymbol{r}_N = \boldsymbol{r}_f, \quad \boldsymbol{v}_N = \boldsymbol{v}_f. \tag{13}$$

Dynamics (trapezoidal with Simpson correction). For $i = 1, \ldots, N-1$,

$$\boldsymbol{v}_{i+1} = \boldsymbol{v}_i + \Delta t\, \boldsymbol{g} + \tfrac{\Delta t}{2}\big(\boldsymbol{u}_i + \boldsymbol{u}_{i+1}\big), \tag{14}$$

$$\boldsymbol{r}_{i+1} = \boldsymbol{r}_i + \tfrac{\Delta t}{2}\big(\boldsymbol{v}_i + \boldsymbol{v}_{i+1}\big) + \tfrac{\Delta t^2}{12}\big(\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\big). \tag{15}$$

Mass (trapezoidal). For $i = 1, \ldots, N-1$,

$$z_{i+1} = z_i - \tfrac{\alpha\,\Delta t}{2}\big(s_i + s_{i+1}\big). \tag{16}$$

Thrust magnitude (SOC). For $i = 1, \ldots, N$,

$$\|\boldsymbol{u}_i\| \le s_i. \tag{17}$$

Taylor bounds on s. Let $z_{0,i}$ be a reference value of z corresponding to the *maximum–thrust* mass history and $z_{1,i}$ the *minimum–thrust* one:

$$z_{0,i} := \ln\!\big(m_{\text{wet}} - \alpha\,\rho_2\,(i-1)\Delta t\big), \tag{18}$$

$$z_{1,i} := \ln\!\big(m_{\text{wet}} - \alpha\,\rho_1\,(i-1)\Delta t\big), \tag{19}$$

$$\mu_{1,i} := \frac{\rho_1}{m_{\text{wet}} - \alpha\,\rho_2\,(i-1)\Delta t} = \rho_1\, e^{-z_{0,i}}, \tag{20}$$

$$\mu_{2,i} := \frac{\rho_2}{m_{\text{wet}} - \alpha\,\rho_2\,(i-1)\Delta t} = \rho_2\, e^{-z_{0,i}}. \tag{21}$$

The convex lower and linear upper bounds on s_i are

$$s_i \ge \mu_{1,i}\Big[1 - (z_i - z_{0,i}) + \tfrac{1}{2}(z_i - z_{0,i})^2\Big], \tag{22}$$

$$s_i \le \mu_{2,i}\Big[1 - (z_i - z_{0,i})\Big], \tag{23}$$

together with the physical envelope on z,

$$z_{0,i} \le z_i \le z_{1,i}. \tag{24}$$

Inequality (22) is a conservative under–approximation of ρ_1/m (quadratic lower bound of a convex function gives a convex constraint); (23) is an over–approximation of ρ_2/m by its tangent line at $z_{0,i}$ (a linear upper bound of a convex function is convex). Together they keep the feasible set convex while tightly tracking the true bounds (10).

Subsurface constraint (active). To forbid flight below the landing pad,

$$r_i^{(2)} \ge -1 \qquad (i = 1, \ldots, N), \tag{25}$$

with $r^{(2)}$ the vertical component. (The $-1\,\text{m}$ margin is the value hard–coded in the script.)

4 Optional constraints (disabled in `p4.m`)

Two additional convex constraints are present in the source but commented out. They are included in the formulation here for completeness.

Thrust–pointing cone. Restricting the thrust vector to lie within a half–cone of aperture θ around the vertical,

$$u_i^{(2)} \geq s_i \cos\theta \qquad (i = 1, \ldots, N). \tag{26}$$

Glide–slope constraint. Keeping the vehicle inside a cone of slope γ anchored at the landing site,

$$r_i^{(1)} \leq \frac{r_i^{(2)}}{\tan\gamma} \qquad (i = 1, \ldots, N). \tag{27}$$

In `p4.m` these are the lines that would activate them, but they are commented out in the current run.

5 MATLAB/CVX implementation walkthrough

The entire program lives in a single self–contained script `p4.m`. It has no dependencies on the other MATLAB files in this repository except for `plot_run2D.m`, which is called at the end only to visualize the results.

5.1 Parameter block

Lines 14–35 declare the numeric values listed in Table 1 and pre–compute the constants α, ρ_1 and ρ_2 defined by (3)–(4):

```
alpha = 1 / (Isp * g0 * cosd(phi));
r1 = min_throttle * T_max * cosd(phi);
r2 = max_throttle * T_max * cosd(phi);
```

5.2 Decision variables and objective

Inside the CVX block, five trajectory arrays are declared and z_N is chosen as the objective:

```
cvx_solver SEDUMI
cvx_begin
    variables r(2,N) v(2,N) u(2,N) z(1,N) s(1,N)
    maximize( z(N) )
    subject to
        ...
cvx_end
```

5.3 Constraint loops

The constraints are split across two `for`–loops:

- A **dynamics loop** over $i = 1, \ldots, N-1$ encoding (14), (15) and (16).

- A **pointwise loop** over $i = 1, \ldots, N$ encoding the SOC constraint (17), the Taylor bounds (22)–(23), and the mass envelope (24). Inside this loop the reference quantities $z_{0,i}$, $z_{1,i}$, $\mu_{1,i}$ and $\mu_{2,i}$ are recomputed every iteration from the current index i.

The boundary conditions and the subsurface constraint (25) are vectorized one–liners outside the loops. The pointing (26) and glide–slope (27) constraints are present as commented–out lines.

5.4 Post–processing

After `cvx_end`, the script recovers the mass profile $m_i = e^{z_i}$ and hands the result to the plotting helper:

```
m_vals = exp(z);
plot_run2D(tv, r, v, u, m_vals);
```

`plot_run2D` produces six figures: position vs. time, velocity vs. time, the 2D trajectory with thrust quiver, the per–axis acceleration, the thrust magnitude, and the acceleration magnitude.

6 How to run

1. Install **MATLAB** and the **CVX** modeling toolbox [3] with the SeDuMi solver enabled (`cvx_solver SEDUMI`).

2. From the repository root, run

```
>> p4
```

3. Expected output: the CVX solver log, followed by six figure windows generated by `plot_run2D`. The optimal trajectory reproduces Figure 6 of [1].

7 Extension to three dimensions

This section documents how the convex formulation of §3 extends to a full three–dimensional landing problem. No 3D implementation is shipped in the current repository; the material below specifies the mathematical form to be used by a future `p4_3D.m`.

7.1 Coordinate conventions

Let $\boldsymbol{r}(t), \boldsymbol{v}(t), \boldsymbol{u}(t) \in \mathbb{R}^3$, with the convention that $\boldsymbol{e}_3$ points vertically upward. The Mars gravity vector becomes

$$\boldsymbol{g} = (0,\ 0,\ -3.7114)\ \mathrm{m/s}^2. \tag{28}$$

Note that Acikmese & Ploen [1] use $\boldsymbol{e}_1$ as the radial (vertical) axis; this note adopts the engineering convention $\boldsymbol{e}_3$ instead so that the 3D code can be obtained from `p4.m` by promoting the vertical slot consistently (the 2D script treats `r(2,:)` as vertical).

7.2 Continuous–time dynamics

The change of variables of §2.2 carries over unchanged; only the translational state is promoted to $\mathbb{R}^3$, while the logarithmic mass z and the thrust slack s remain scalar:

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \qquad \dot{\boldsymbol{v}} = \boldsymbol{u} + \boldsymbol{g}, \tag{29}$$

$$\dot{z} = -\alpha\, s, \qquad \|\boldsymbol{u}\| \le s. \tag{30}$$

The lossless convexification argument and the Taylor bounds (22)–(23) on s are dimension–independent and apply verbatim.

7.3 Discretized 3D convex program

Decision variables on the same N knot points as in §3 are

$$\boldsymbol{r}_i, \boldsymbol{v}_i, \boldsymbol{u}_i \in \mathbb{R}^3, \qquad z_i, s_i \in \mathbb{R}, \qquad i = 1, \ldots, N.$$

The objective (max z_N), the boundary conditions, the trapezoidal dynamics (14)–(15), the mass update (16), the Taylor bounds (22)–(23), and the mass envelope (24) are identical in form; only the vector variables now carry a third component.

Thrust magnitude (SOC, $\mathbb{R}^3$).

$$\|\boldsymbol{u}_i\| \le s_i \qquad (i = 1, \ldots, N), \tag{31}$$

a 3D second–order cone directly supported by CVX/SeDuMi.

Subsurface constraint.

$$r_i^{(3)} \ge -1 \qquad (i = 1, \ldots, N). \tag{32}$$

Thrust–pointing cone. A half–cone of aperture θ around the vertical becomes

$$u_i^{(3)} \ge s_i \cos\theta \qquad (i = 1, \ldots, N). \tag{33}$$

Axisymmetric glide–slope cone (SOC). Keeping the vehicle inside an axisymmetric cone of slope γ anchored at the landing site generalizes (27) to

$$\sqrt{\left(r_i^{(1)}\right)^2 + \left(r_i^{(2)}\right)^2} \;\le\; \frac{r_i^{(3)}}{\tan\gamma} \qquad (i = 1, \ldots, N), \tag{34}$$

which is a second–order cone constraint and thus convex.

7.4 Example 3D parameters

Table 2 lists a suggested 3D initial/terminal state; the remaining vehicle and discretization parameters are inherited from Table 1 to permit direct comparison with the 2D run.

7.5 Remarks

Because the lossless convexification structure is dimension–independent, the 3D program is still a standard SOCP and can be solved by SeDuMi with only a modest increase in problem size ($3N$ additional primal variables versus the 2D case). Visualization of the 3D trajectory is out of scope here and would require a companion script `plot_run3D.m` replacing `plot_run2D.m`.

Symbol	Meaning	Value
$\boldsymbol{r}_0$	initial position	$(2000,\ 500,\ 1500)\,\mathrm{m}$
$\boldsymbol{v}_0$	initial velocity	$(100,\ 50,\ -75)\,\mathrm{m/s}$
$\boldsymbol{r}_f, \boldsymbol{v}_f$	terminal state	$\mathbf{0},\ \mathbf{0}$
$\boldsymbol{g}$	Mars gravity vector	$(0,\ 0,\ -3.7114)\,\mathrm{m/s}^2$
θ	pointing half–angle	50°
γ	glide–slope angle	4°

Table 2: Suggested 3D boundary conditions and cone angles; all other parameters $(m_{\mathrm{dry}}, m_{\mathrm{wet}}, I_{sp}, T_{\max}, \eta_{\min}, \eta_{\max}, \phi, t_f, \Delta t, N)$ are taken from Table 1. These values are indicative and are not produced by any script in the current repository.

References

[1] B. Acikmese and S. R. Ploen. Convex programming approach to powered descent guidance for Mars landing. *Journal of Guidance, Control, and Dynamics*, 30(5):1353–1366, 2007.

[2] B. Acikmese, L. Blackmore, D. P. Scharf, and A. Wolf. Enhancements on the convex programming based powered descent guidance algorithm for Mars landing. In *AIAA/AAS Astrodynamics Specialist Conference*, 2008.

[3] M. Grant and S. Boyd. CVX: MATLAB software for disciplined convex programming. `http://cvxr.com/cvx/`.